Chapter 2: Electrochemistry

Formula Sheet + Tips & Tricks

Formula Sheet

I. Cell Potential & EMF

EMF of a Cell

$$\text{EMF} = E_C - E_A$$

E_C = electrode potential of cathode; E_A = electrode potential of anode. Standard values assume 1 molal conc., 1 atm pressure, 298 K.

Nernst Equation (natural log form)

$$E = E^\circ - \frac{2.303RT}{nF}\log_{10}\frac{[\text{product}]}{[\text{reactant}]}$$

$R = 8.314$ J/mol/K; $F = 96500$ C; $T = 298$ K; n = moles of electrons transferred.

Nernst Equation (298 K simplified)

$$E = E^\circ - \frac{0.059}{n}\log_{10}\frac{[\text{product}]}{[\text{reactant}]}$$

Simplified form obtained by substituting R, T, F at 298 K into the Nernst equation.

Single Electrode Potential

$$E = E^\circ + \frac{0.0591}{n}\log(\text{ion concentration})$$

For reduction at a single electrode; note the + sign (opposite convention to the full-cell Nernst equation).

II. Gibbs Energy & Equilibrium

Work Done by Cell

$$W = nFE^\circ$$

Work done = charge × potential; 1 mole of electrons carries charge F (96500 C).

Gibbs Free Energy and EMF

$$\Delta G = -nFE^\circ_{\text{cell}}$$

Negative value indicates a spontaneous reaction; n = moles of electrons, $F = 96500$ C/mol.

Gibbs Energy at Equilibrium

$$\Delta G = -2.303\,RT\log K_c$$

Relates standard free energy change to the equilibrium constant K_c.

Gibbs-Helmholtz Equation

$$\Delta G = \Delta H - T\Delta S$$

Used to test spontaneity and transition temperature; set $\Delta G = 0$ to solve for the temperature at which a reaction becomes non-spontaneous.

III. Conductance

Resistance and Conductance

$$R = \frac{V}{I}, \qquad C = \frac{1}{R}$$

R = resistance (ohm); C = conductance (siemens).

Specific Resistance / Cell Constant

$$\rho = R\cdot\frac{A}{\ell}, \qquad \frac{\ell}{A} = \text{cell constant} = \kappa \times R$$

ℓ = distance between electrodes; A = area of cross-section.

Conductivity (Specific Conductance)

$$\kappa = \frac{1}{\rho} = \frac{1}{R}\times\frac{\ell}{A} = \text{conductance} \times \text{cell constant}$$

Symbol κ (kappa); units S cm^{-1}.

Molar Conductivity

$$\Lambda_m = \frac{\kappa \times 1000}{\text{molarity}} = \kappa \times V$$

V = volume (in mL containing 1 mole solute) of solution.

Equivalent Conductivity

$$\Lambda_{eq} = \frac{\kappa \times 1000}{\text{normality}}$$

Normality $= \dfrac{\text{given mass}}{\text{equivalent mass}}$; eq. mass $= \dfrac{\text{molar mass}}{\text{valency}}$.

Kohlrausch's Law

$$\Lambda_m^\circ = \nu_+\lambda_+^\infty + \nu_-\lambda_-^\infty$$

Molar conductivity at infinite dilution is the sum of ionic contributions.

Degree of Dissociation from Conductance

$$\alpha = \frac{\Lambda_m(\text{given})}{\Lambda_m^\circ(\text{calculated})}$$

Ratio of molar conductivity at given concentration to that at infinite dilution.

IV. Faraday's Laws of Electrolysis

Charge Passed

$$Q = I \times t$$

I = current (amperes); t = time (seconds).

Faraday's First Law

$$m = Z\,Q = \frac{Z\,I\,t}{1}$$

m = mass deposited; Z = electrochemical equivalent.

Mass Deposited (combined form)

$$m = \frac{E\,I\,t}{nF}$$

E = equivalent weight; n = valency/electrons transferred; F = 96500 C/mol.

Tips, Tricks & Hacks

- **Idea for identifying cathode/anode**: the electrode with the greater (more positive) standard reduction potential is always the cathode.
- In a **concentration cell**, $E^\circ_{\text{cell}} = 0$ since both electrodes are the same element; only the Nernst-equation log term contributes to the EMF. The more dilute electrode is the anode, the more concentrated is the cathode.
- Constants to memorise: $R = 8.314$ J/mol/K, $F = 96500$ C, $T = 298$ K (standard conditions).
- At equilibrium, $E_{\text{cell}} = 0$ and $Q = K_c$ – this is the bridge between the Nernst equation and the ΔG-K_c relation.
- Watch the sign convention: full-cell Nernst equation uses $E^\circ - \dfrac{0.059}{n}\log Q$, but single-electrode (reduction) potential uses $E^\circ + \dfrac{0.0591}{n}\log[\text{ion}]$.